PROJECT REPORT

ON

KIIT CAMPUS NAVIGATOR USING A SEARCH ALGORITHM*

Submitted in partial fulfillment for the course
Artificial Intelligence Lab

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ABSTRACT

The KIIT Campus Navigator is a web-based application designed to assist users in finding the shortest path between different locations within the KIIT University campus. Large university campuses often create navigation difficulties for new students and visitors. This project aims to provide an efficient and user-friendly solution by implementing Artificial Intelligence search techniques.

The system is developed using Python and the Flask framework and utilizes the A* (A-Star) search algorithm to determine the optimal route between selected source and destination points. The campus map is represented as a graph where various campus locations act as nodes and the connecting paths act as edges with associated distances. A heuristic function based on geographical coordinates is used to estimate the shortest possible path, improving the efficiency of the search process.

The application provides an interactive web interface where users can select their starting and ending locations and obtain the shortest path along with distance information. The project demonstrates the practical implementation of Artificial Intelligence in solving real-world navigation problems and highlights the effectiveness of heuristic search algorithms like A*.

This system can be further enhanced by integrating real-time navigation, GPS tracking, and mobile application support to provide an advanced smart campus navigation solution in the future.

INTRODUCTION

The rapid growth of large educational institutions and university campuses has made navigation within campus premises increasingly challenging for students, faculty members, and visitors. New students and visitors often face difficulty in locating different departments, academic blocks, hostels, administrative offices, and other essential facilities within a large campus. Traditional static maps and signboards are often insufficient to provide efficient navigation, leading to confusion and time consumption.

With the advancement of technology and Artificial Intelligence, intelligent navigation systems can be developed to provide accurate and efficient pathfinding solutions. Artificial Intelligence-based search algorithms are widely used in real-world applications such as GPS navigation, robotics, and game development to determine optimal paths between locations. Implementing such algorithms in a campus environment can significantly improve navigation efficiency and user experience.

The KIIT Campus Navigator is an AI-based web application designed to determine the shortest path between various locations within the university campus. The system utilizes the A* (A-Star) search algorithm, a widely used heuristic search algorithm known for its efficiency in pathfinding and graph traversal. By representing campus locations as nodes and pathways as edges in a graph structure, the system calculates the most optimal route between the selected source and destination.

The project is developed using Python and the Flask web framework, providing an interactive and user-friendly interface for users. The integration of heuristic functions based on geographical coordinates enhances the performance of the pathfinding process. This project demonstrates the practical implementation of Artificial Intelligence concepts in solving real-world navigation problems and highlights the importance of intelligent systems in modern smart campus environments.

PROBLEM STATEMENT

The rapid expansion of large university campuses often creates difficulty for students, faculty members, and visitors in locating different academic and administrative facilities efficiently. Traditional navigation methods such as static maps and signboards are often insufficient and time-consuming, especially for new students and visitors who are unfamiliar with the campus layout.

There is a need for an intelligent system that can provide accurate and efficient navigation within the campus environment. The system should be capable of determining the shortest and most optimal path between different campus locations while considering distance and connectivity. Therefore, the problem addressed in this project is to design and develop an AI-based campus navigation system that utilizes efficient search algorithms to help users easily navigate between various campus locations.

OBJECTIVES

The main objectives of the KIIT Campus Navigator project are:

• To design and develop an AI-based campus navigation system.
• To implement the A* (A-Star) search algorithm for optimal pathfinding.
• To represent campus locations as nodes and pathways as edges in a graph structure.
• To calculate the shortest path between selected source and destination locations.
• To develop a user-friendly web interface using Flask.
• To improve navigation efficiency within the campus environment.
• To demonstrate the practical application of Artificial Intelligence in solving real-world problems.

SYSTEM ARCHITECTURE

The system architecture of the KIIT Campus Navigator is designed to provide an efficient and user-friendly navigation experience within the university campus. The system follows a client-server architecture where the user interacts with a web interface, and the backend processes the navigation requests using Artificial Intelligence algorithms.

The user provides input by selecting the source and destination locations through the web interface. The request is then sent to the Flask-based backend server, where the campus map is represented as a graph consisting of nodes and edges. Each node represents a campus location, and the edges represent the paths connecting them with associated distances.

The frontend map interface is implemented using Leaflet.js with OpenStreetMap integration for visualization of campus locations.

The A* search algorithm is implemented in the backend to determine the shortest path between the selected locations. A heuristic function based on geographical coordinates is used to estimate the distance between nodes and improve the efficiency of the pathfinding process. The algorithm processes the input data and calculates the optimal route.

After processing, the system sends the result back to the user interface, where the shortest path and distance information are displayed. The architecture ensures efficient communication between the user interface and the backend processing unit, providing accurate and fast navigation results.

ALGORITHM USED

The KIIT Campus Navigator utilizes the A* (A-Star) search algorithm to determine the shortest path between selected campus locations. A* is an informed search algorithm widely used in pathfinding and graph traversal problems due to its efficiency and accuracy.

The A* algorithm calculates the optimal path using the evaluation function:

$f(n) = g(n) + h(n)$

where:

$g(n)$ represents the actual cost from the starting node to the current node, $h(n)$ represents the heuristic estimated cost from the current node to the destination, and $f(n)$ represents the total estimated cost of the path through the current node.

In this project, the campus map is represented as a graph consisting of nodes and edges. Each node represents a campus location, and the edges represent paths with associated distances. The heuristic function is calculated using geographical coordinates (latitude and longitude) to estimate the straight-line distance between two locations.

The algorithm works by maintaining an open list of nodes to be evaluated and a closed list of nodes that have already been evaluated. At each step, the node with the lowest $f(n)$ value is selected for expansion. The algorithm continues evaluating neighboring nodes and updating their cost values until the destination node is reached. This ensures that the shortest and most efficient path is determined.

The use of the A* algorithm significantly improves navigation efficiency by combining actual path cost with heuristic estimation, making it suitable for real-time navigation systems such as the KIIT Campus Navigator.

TIME COMPLEXITY

The time complexity of the A* search algorithm depends on the number of nodes and edges in the graph. In the worst-case scenario, the time complexity of the A* algorithm is similar to that of Dijkstra's algorithm and can be expressed as O(E log V), where E represents the number of edges and V represents the number of vertices in the graph.

The use of a heuristic function in the A* algorithm helps reduce the number of nodes explored, thereby improving efficiency compared to uninformed search algorithms. By guiding the search toward the destination, the algorithm ensures faster computation of the shortest path in most practical scenarios.

ADVANTAGES

- Provides efficient and accurate navigation within the campus.
- Helps new students and visitors easily locate different campus facilities.
- Uses A* search algorithm for optimal pathfinding.
- Reduces time required to find routes between locations.
- User-friendly web interface for easy interaction.
- Demonstrates practical implementation of Artificial Intelligence.
- Can be extended to larger smart campus systems.

LIMITATIONS

- The system works only for predefined campus locations.
- It does not include real-time traffic or crowd data.
- Navigation is limited to mapped campus areas only.
- Requires internet connection to run the web application.
- Newly developed campuses not available on digital maps cannot be included currently.

FUTURE SCOPE

• Integration of real-time navigation and live traffic updates within the campus.
• Addition of GPS-based tracking for dynamic route calculation.
• Development of a mobile application version for wider accessibility.
• Integration with smart campus systems for automated guidance.
• Inclusion of newly developed campuses such as KIIT Campuses 21–24 once accurate geographical coordinates and digital map data become available.
• Enhancement of the heuristic model using advanced AI techniques for improved performance.

RESULT AND OUTPUT

The KIIT Smart Campus Navigator system was successfully implemented and tested using various campus locations. The application allows users to select a starting point and destination within the KIIT campus and calculates the shortest path using the A* search algorithm.

The system displays the optimal path along with the calculated distance between selected campuses. The results confirm that the algorithm efficiently determines the shortest route based on predefined campus coordinates and graph representation.

1. Shortest Path Result (Campus 6 → Campus 25)

- Distance: 0.4 km
- Nodes Explored: 2

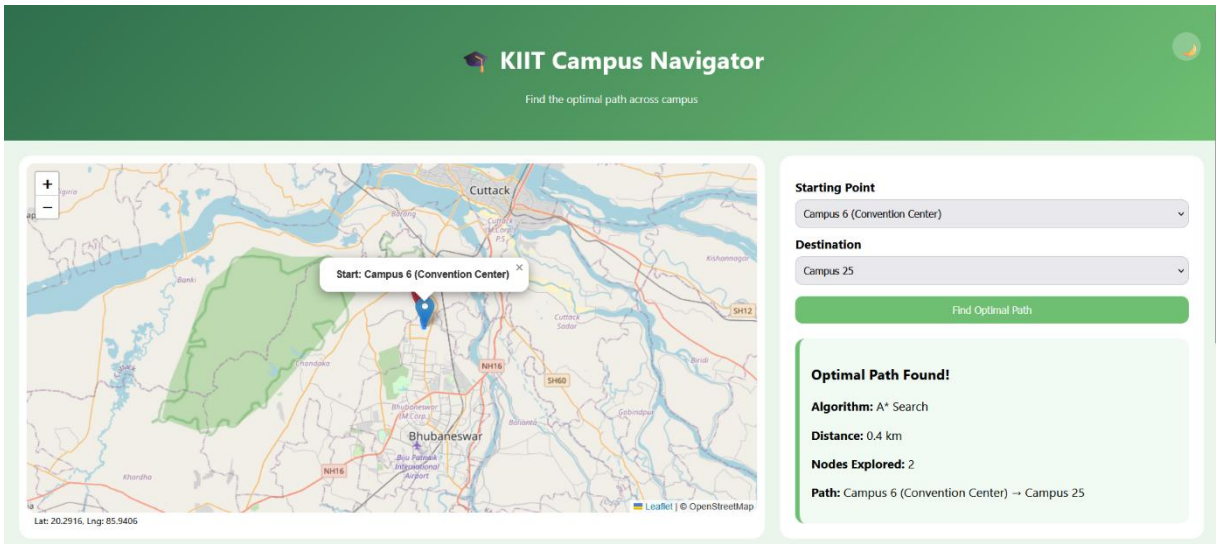

Figure 1: Shortest path between Campus 6 and Campus 25

2. Additional Path Result (Campus 6 → Campus 15)

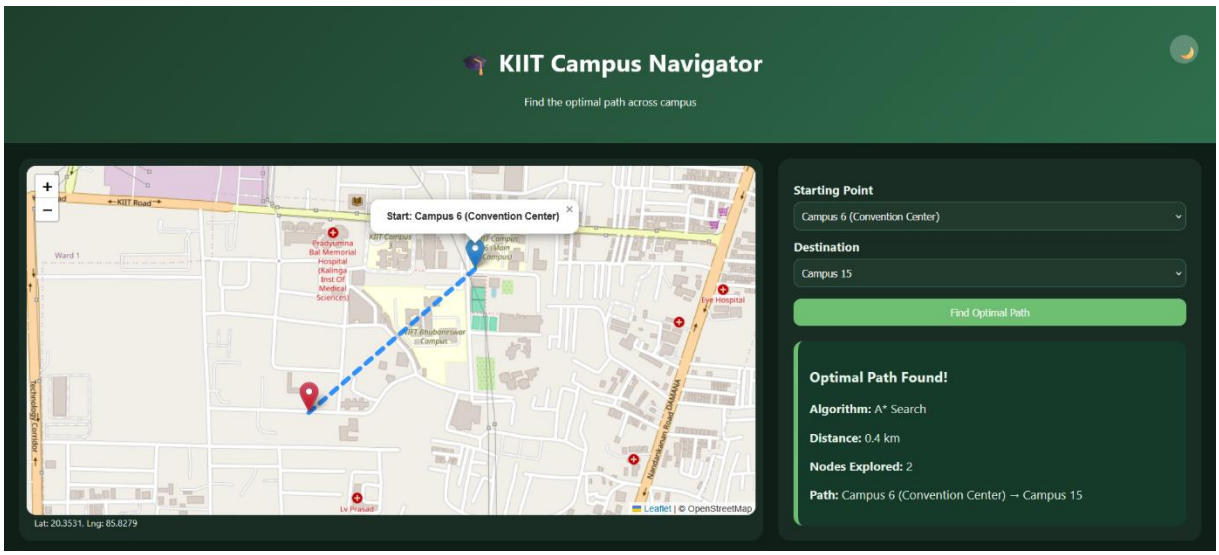

Figure 2: Shortest path between Campus 6 and Campus 15

3. Feedback Module

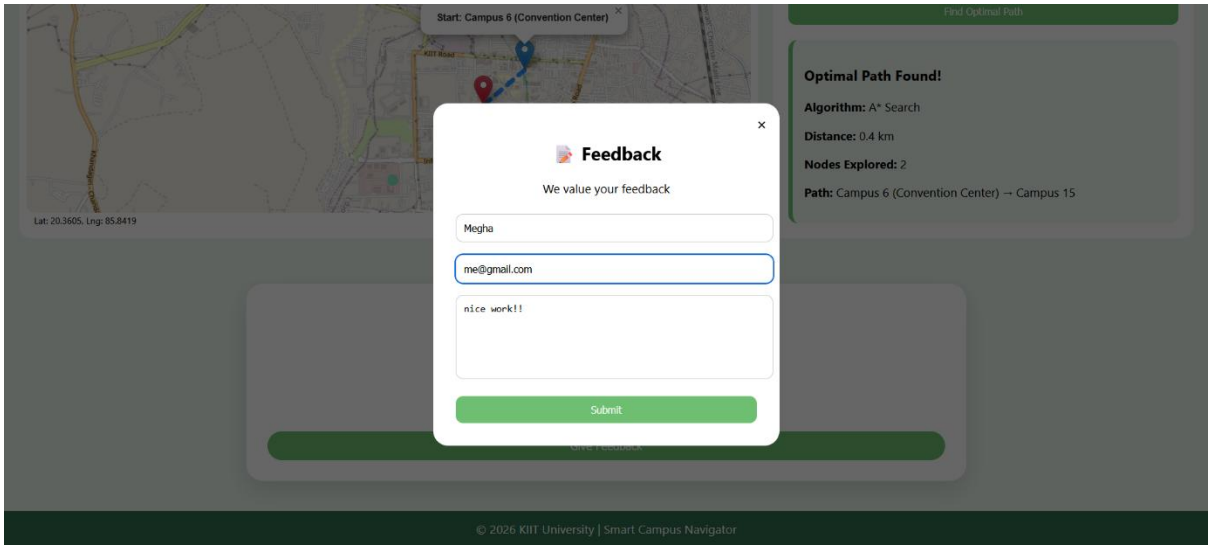

Figure 3: Feedback module interface

The following screenshots show the working output of the KIIT Campus Navigator system, including shortest path calculation and feedback module interface.

CONCLUSION

The KIIT Campus Navigator successfully demonstrates the practical implementation of Artificial Intelligence in solving real-world navigation problems within a university campus. By utilizing the A* search algorithm and heuristic techniques, the system efficiently determines the shortest path between various campus locations and provides accurate navigation results through a user-friendly web interface.

This project highlights the importance of intelligent systems in modern smart campus environments and showcases how AI-based algorithms can improve navigation efficiency and user experience. The integration of graph-based representation and heuristic search ensures optimal pathfinding and effective system performance.

Overall, the project serves as a valuable application of Artificial Intelligence concepts and can be further enhanced with advanced features such as real-time tracking, mobile integration, and expanded campus coverage in the future.